Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-II Physics Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Secondary School Certificate (SSC) Part II Physics. It contains comments on candidates' responses to the 2019 SSC Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

This year, candidates performed well on questions related problems solving by applying the relation $f = \frac{1}{T}$ and V= f λ, refraction of light, total internal reflection, mutual induction, process of fission, use of formula $T = 2\pi\sqrt{\frac{l}{g}}$ for problems related to simple pendulum, ray diagrams of a telescope, a long-sighted eye and potential dangers of electrostatic charges. However, candidates did not perform well in questions based on the applications of electrostatic, factors affecting the resistances of a metallic conductor, deflection of electron by magnetic field, echo and also had problems in correctly articulating their answers related to basic definition, reasoning and discussion type questions.

Detailed Comments:

Constructed Response Questions (CRQs)

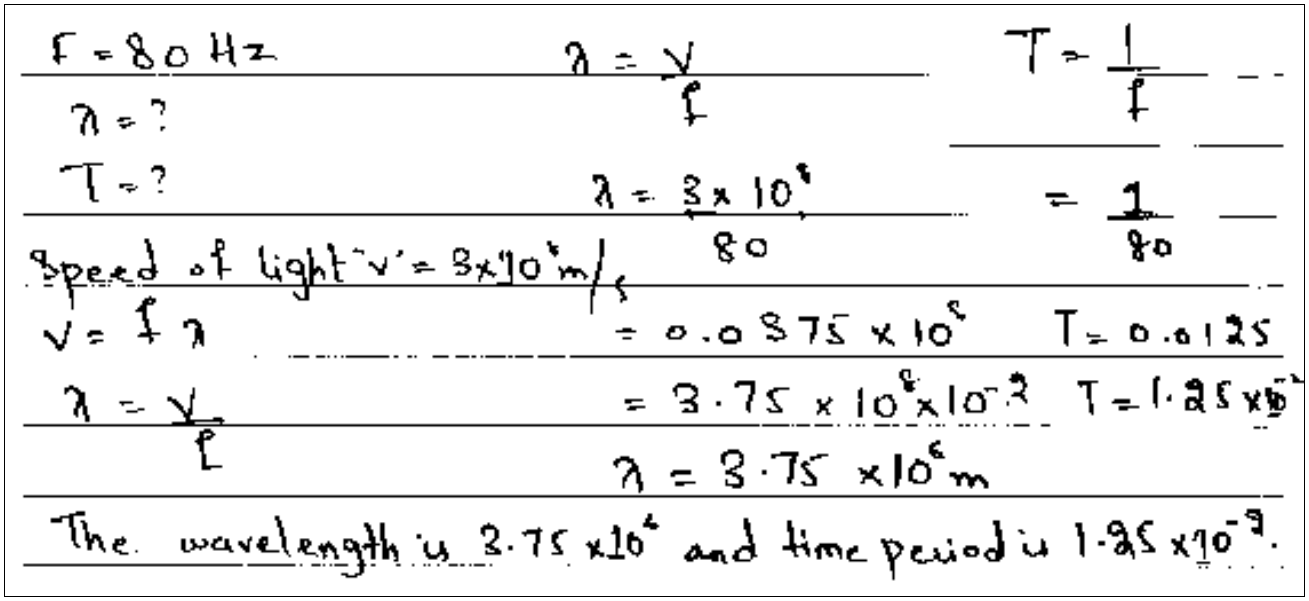
Question 1:

You are given an assignment to operate a radio station broadcasting your school news and activities. If your school allocated a radio wave frequency of FM 80 Hz, then calculate the wavelength and time period of the signal of your school's radio station.

(**Note**: The speed of light in vacuum is $v = 3 \times 10^8$ m/s.)

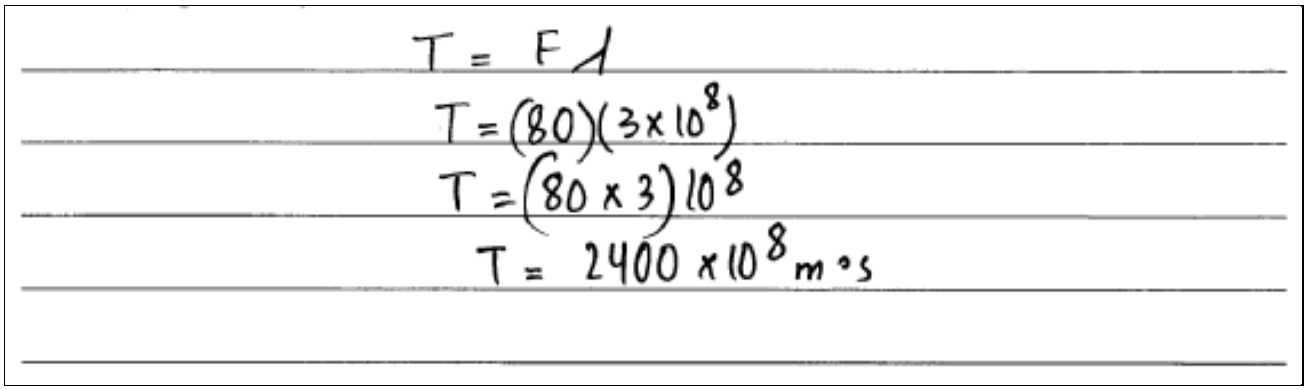
Better responses calculated the correct wavelength and time period. High-scoring candidates used the correct formulae, ($V = f\lambda$ and $T = \frac{1}{f}$) and calculated the correct values of wavelength and time period, i.e. $\lambda = 3.7 \times 10^6$ m and T = 0.0125 s respectively.

Example:



Weaker responses showed candidates' lack of understanding of calculating the correct wavelength and time period. Few candidates were not even able to write the correct formula of velocity and time period as well. They were confused and hence, mixed up two different formulae of velocity and time period and tried to get the correct answer.

Example:



Question 2:

a. Why does a ray of light bend when it crosses the boundary of two media?

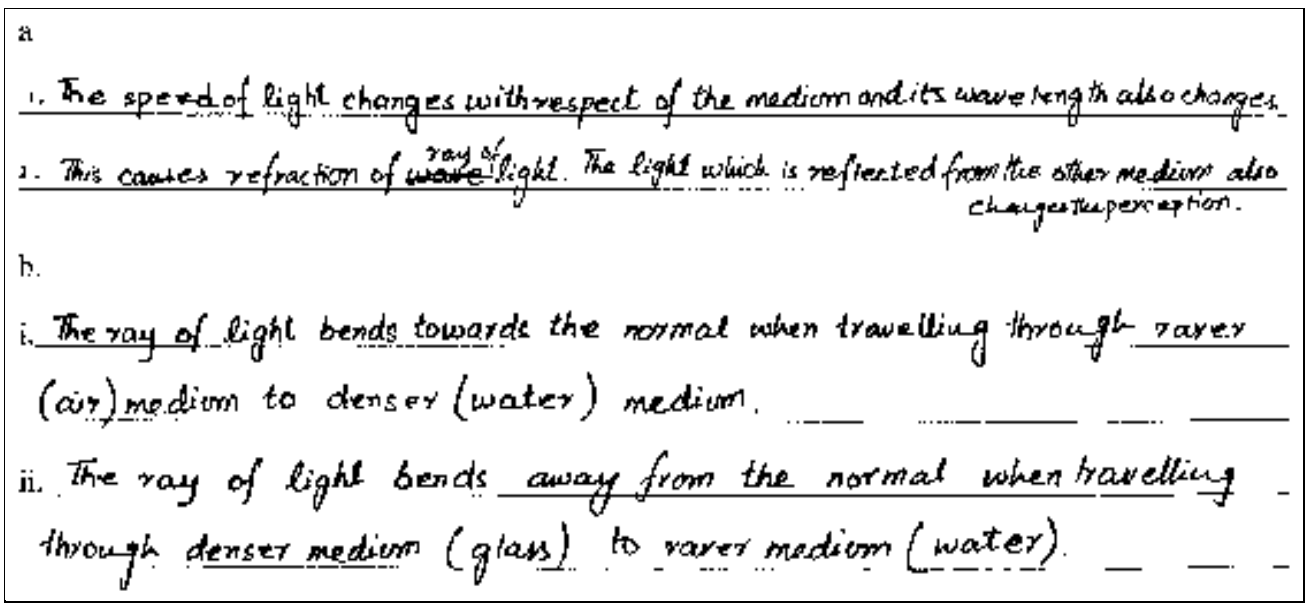
b. In what direction with respect to normal, does a ray of light bend when as it travels through the following media?

   i. From air to water

   ii. From glass to water

Better responses correctly wrote the reason about a ray of light that when it crosses the boundary of two media, it bends. High-scoring candidates gave a reason as 'in a denser medium the speed of light decreases due to the decrease in wavelength. Some of the candidates wrote that 'when a plane-wavelength is incident on the boundary of two media at a certain angle, the distance between wavefronts in denser medium decreases and ultimately bending of light ray occurs'. In part (b), high-scoring candidates correctly indicated the direction of a ray of light when it travelled from air to water and glass to water.

Example:


a

i. The speed of light changes with respect of the medium and its wavelength also changes.

ii. This causes refraction of ~~wave~~ ray of light. The light which is reflected from the other medium also changes the perception.

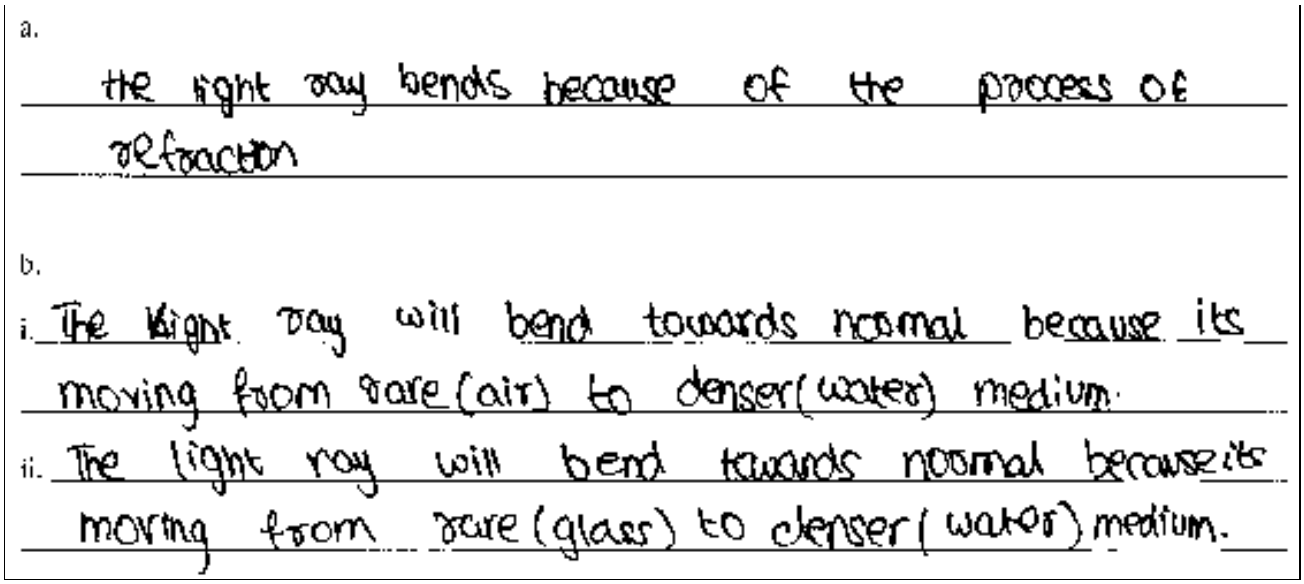
b.

i. The ray of light bends towards the normal when travelling through rarer (air) medium to denser (water) medium.

ii. The ray of light bends away from the normal when travelling through denser medium (glass) to rarer medium (water).

Weaker responses revealed that candidates were unable to give out the correct justification. But in part (b), few of the low-scoring candidates had successfully written the correct direction of the ray of light with respect to the normal, i.e. bends towards the normal and bends away from the normal.

Example:


a.

the light ray bends because of the process of refraction

b.

i. The light ray will bend towards normal because its moving from rare (air) to denser (water) medium.

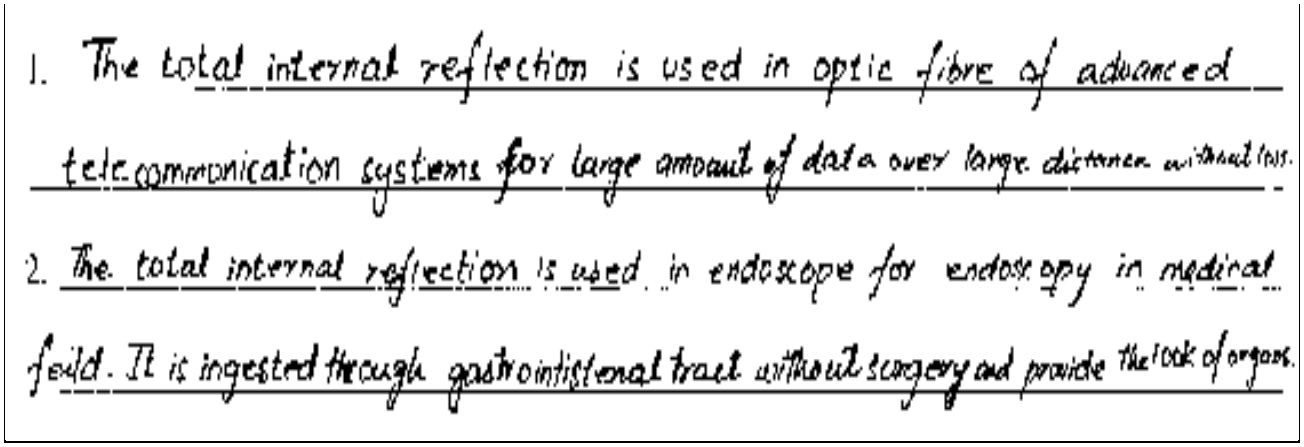
ii. The light ray will bend towards normal because its moving from rare (glass) to denser (water) medium.

Question 3:

Write any TWO uses of total internal reflection in light propagation.

Better responses correctly mentioned two uses of total internal reflection in light propagation. Such as total internal reflection is the operating principle of optical fibres, which are used in endoscopes and telecommunications. The candidates mentioned that it is also the operating principle of automotive rain sensors, which controls automatic windscreen/ windshield wipers.

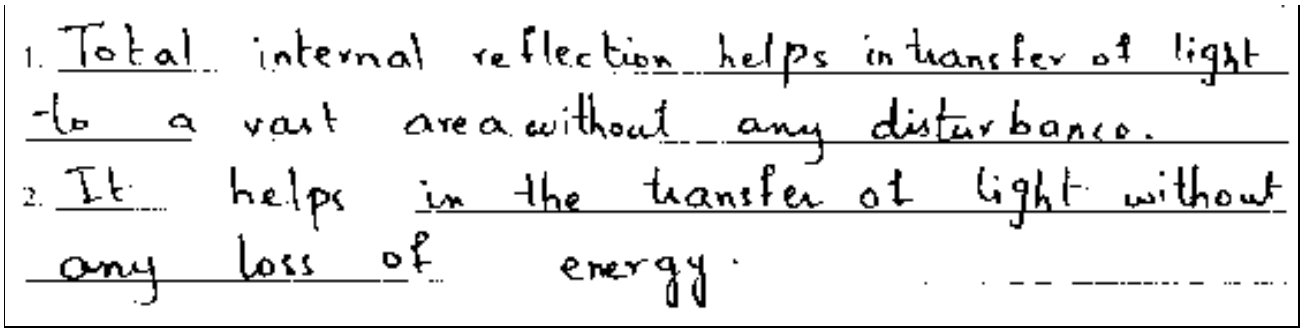
Example:


1. The total internal reflection is used in optic fibre of advanced telecommunication systems for large amount of data over large distance without loss.

2. The total internal reflection is used in endoscope for endoscopy in medical feild. It is ingested through gastrointistenal tract without surgery and provide the look of organs.

Weaker responses showed incorrect uses of total internal reflection. Some of the low-scoring candidates wrote about the concept of critical angle, angle of incidence and angle of refraction. Few candidates incorrectly gave an idea of the light that passes through rare to denser medium.

Example:

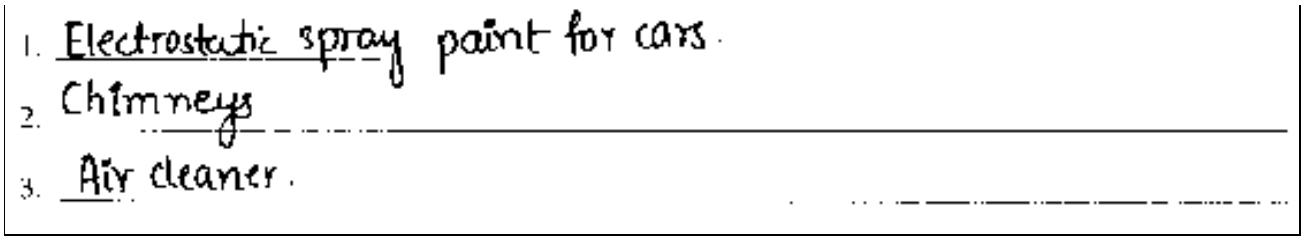


Question 4:

Name any THREE devices that work on the basis of electrostatic phenomena.

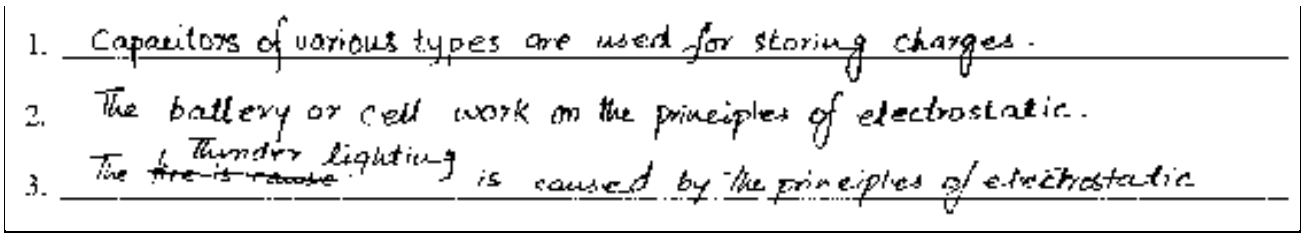
Better responses correctly stated the names of the devices that work on the basis of electrostatic phenomena. High-scoring candidates wrote different names of the devices which included photocopier, laser printers, spray painting, ink jet printers, smoke precipitators, Van de Graff generator and electrostatic air cleaners.

Example:



Weaker responses failed to write the correct names of the devices. Few of the low-scoring candidates wrote names of some electrical devices such as capacitor, battery, resistor etc.

Example:

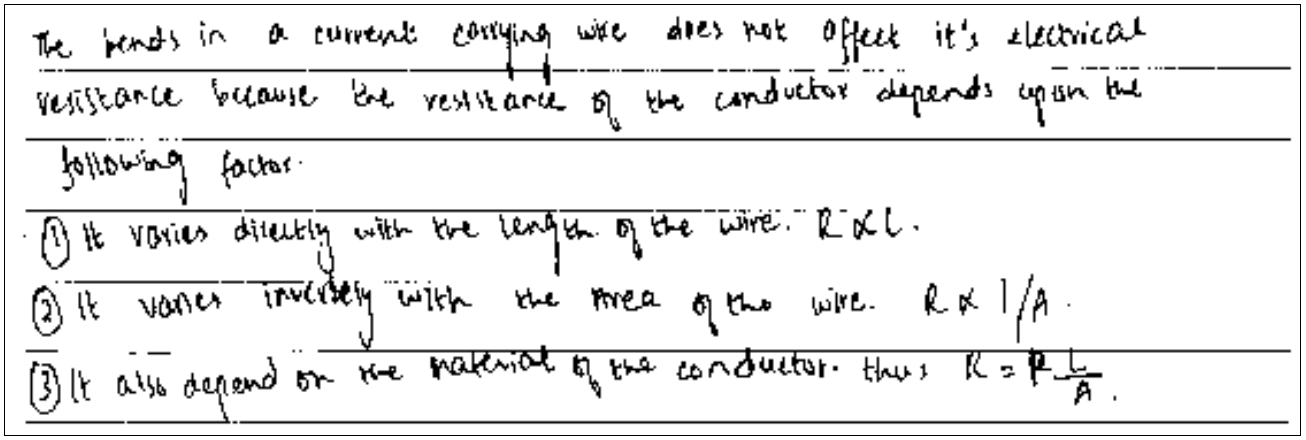


Question 5:

Is it possible that small bends in a current carrying wire affect its electrical resistance? Justify your answer keeping in view the factors affecting electrical resistance.

Better responses stated 'no' and correctly justified their answer with reference to the effect of small bends in a current carrying wire on electrical resistance. High-scoring candidates also wrote the factors upon which the electrical resistance depends such as length, area of cross-section and the nature of the material of the wire. Some of the candidates also wrote the formula, $R=\rho\frac{L}{A}$ in their justification.

Example:


The bends in a current carrying wire does not effect it's electrical resistance because the resistance of the conductor depends upon the following factor.

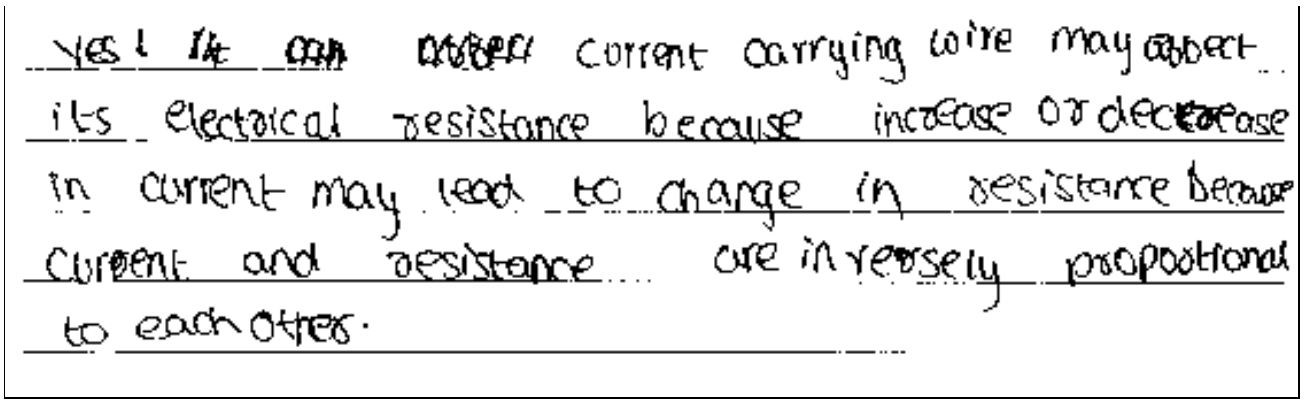
(1) It varies directly with the length of the wire. $R \propto L$.

(2) It varies inversely with the area of the wire. $R \propto 1/A$.

(3) It also depend on the material of the conductor. thus $R = \rho\frac{L}{A}$.

Weaker responses reflected that candidates lack knowledge about the basic concepts of electrical resistance. They wrote that small bends in a current carrying wire affect its electrical resistance. Few candidates also wrote about electricity, conductors and insulators which were irrelevant to the demand of the question.

Example:


Yes! It can effect current carrying wire may effect its electrical resistance because increase or decrease in current may lead to change in resistance because current and resistance are inversely proportional to each other.

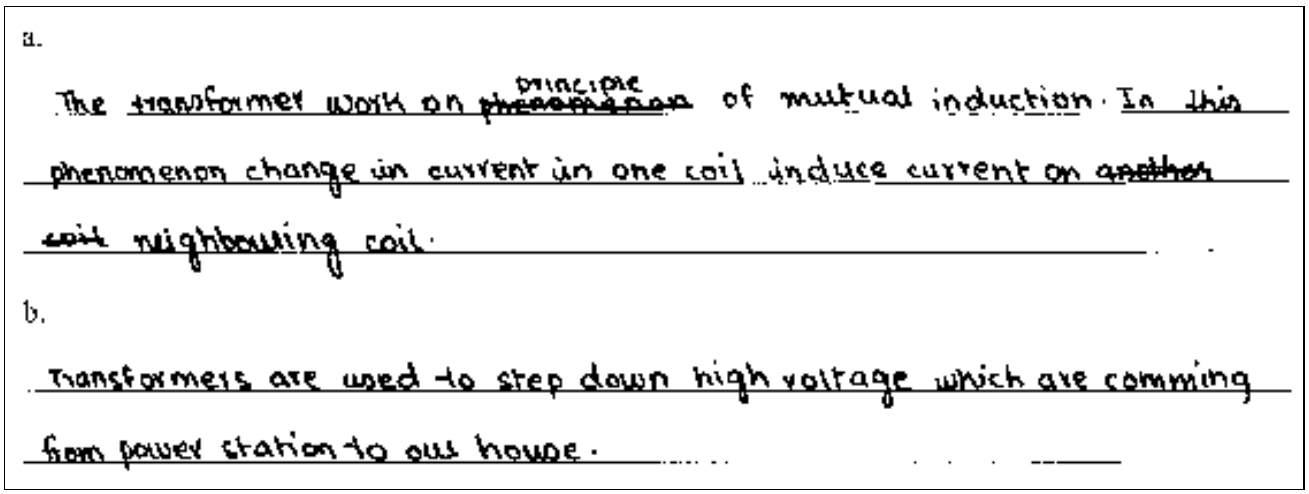
Question 6:

a. Name and define the principle of a transformer on which it works.

b. Write ONE use of a transformer.

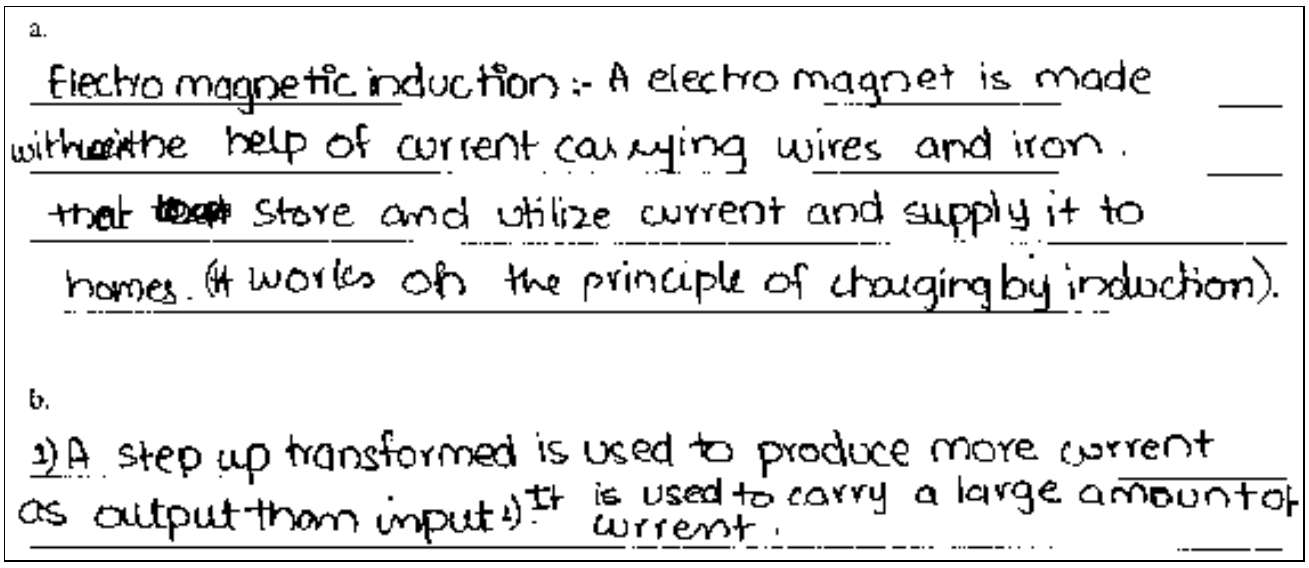
Better responses correctly named and defined the working principle of a transformer. High-scoring candidates in part (a) wrote mutual induction as the name of the principle and defined it as the changing current in one coil induces an electromotive force in another coil. In part (b), they wrote that transformers are used to manipulate the level of voltage flowing through any point in a power grid.

Example:



Weaker responses were unable to name the principle. The incorrect answers include electromagnetic induction and its definition. Some of the low-scoring candidates correctly wrote the use of a transformer such as it step-up or step-down the voltage in a power grid.

Example:

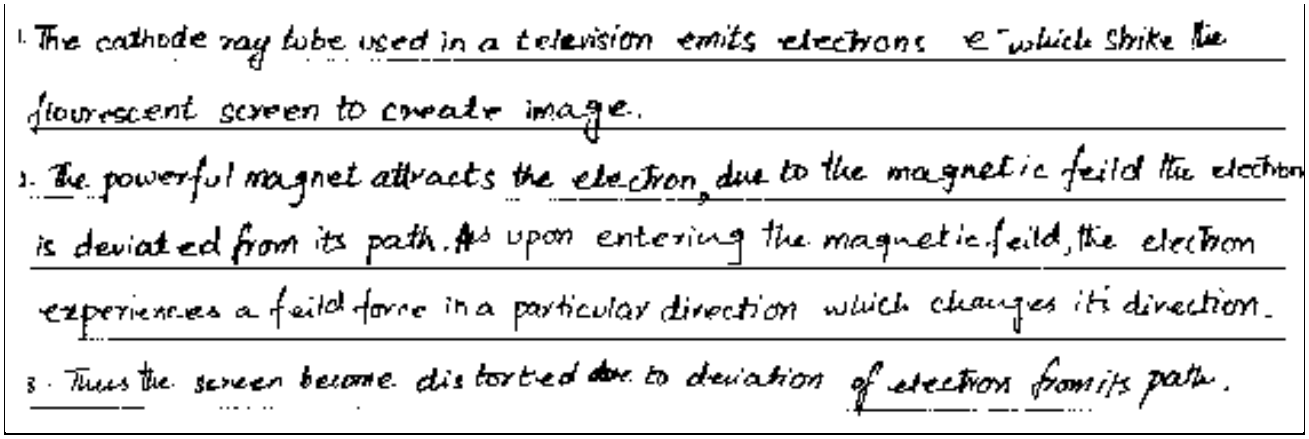


Question 7:

The picture that appears on the screen of a television, that uses cathode rays tube, becomes distorted when a powerful magnet is brought near the screen. Justify the given statement.

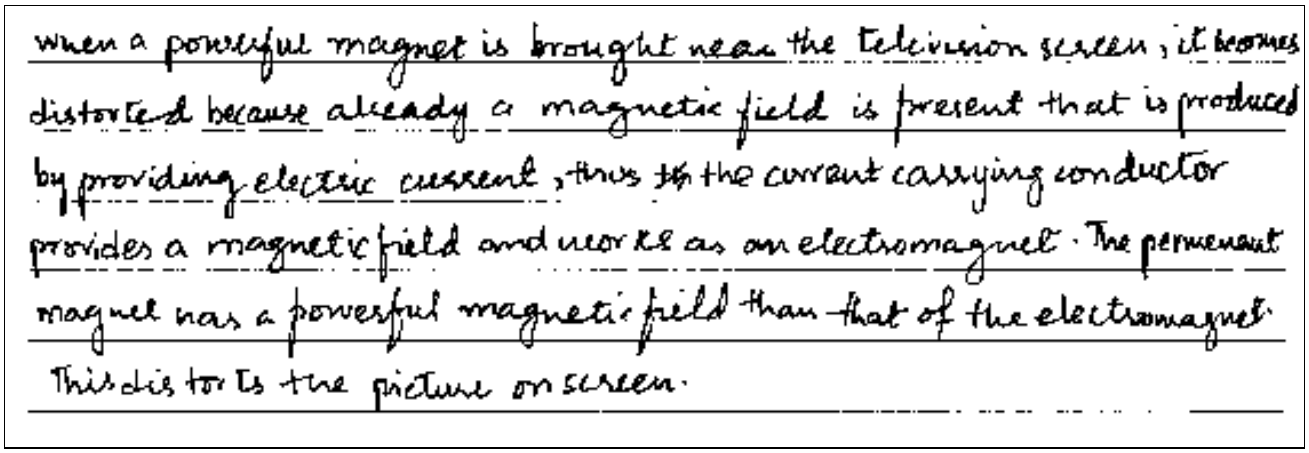
Better responses correctly justified by stating that when a magnet is brought near the screen, charged particles experience a magnetic force that displaces them from their target points on the screen. Thus, the picture is disturbed. Some of the candidates also explained that a magnet distorts the picture as it distorts the path of electrons flowing from the electron gun towards the screen inside the television and as electrons are negatively charged particles; therefore, their motion is disturbed by a magnet.

Example:


1. The cathode ray tube used in a television emits electrons e^- which shike the flourescent screen to create image.
2. The powerful magnet attracts the electron, due to the magnetic feild the electron is deviated from its path. As upon entering the magnetic feild, the electron experiences a feild force in a particular direction which changes its direction.
3. Thus the screen become distorted due to deviation of electron from its path.

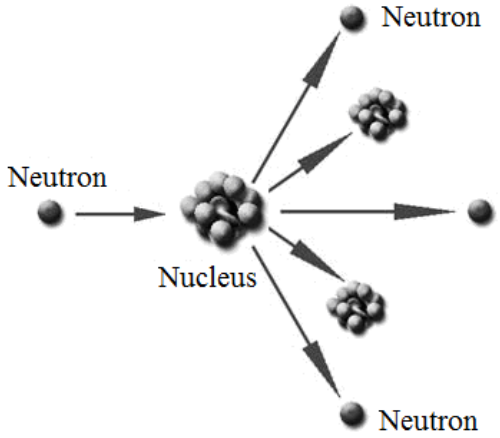
Weaker responses failed to write the correct reason. Few low-scoring candidates wrote about the electric field, magnetic field and electromagnet and tried to explain their answers in terms of charged particles and their movement in cathode rays tube.

Example:


when a powerful magnet is brought near the television screen, it becomes distorted because already a magnetic field is present that is produced by providing electric current, thus the current carrying conductor provides a magnetic field and works as an electromagnet. The permenant magnet has a powerful magnetic field than that of the electromagnet. This distorts the picture on screen.

Question 8:

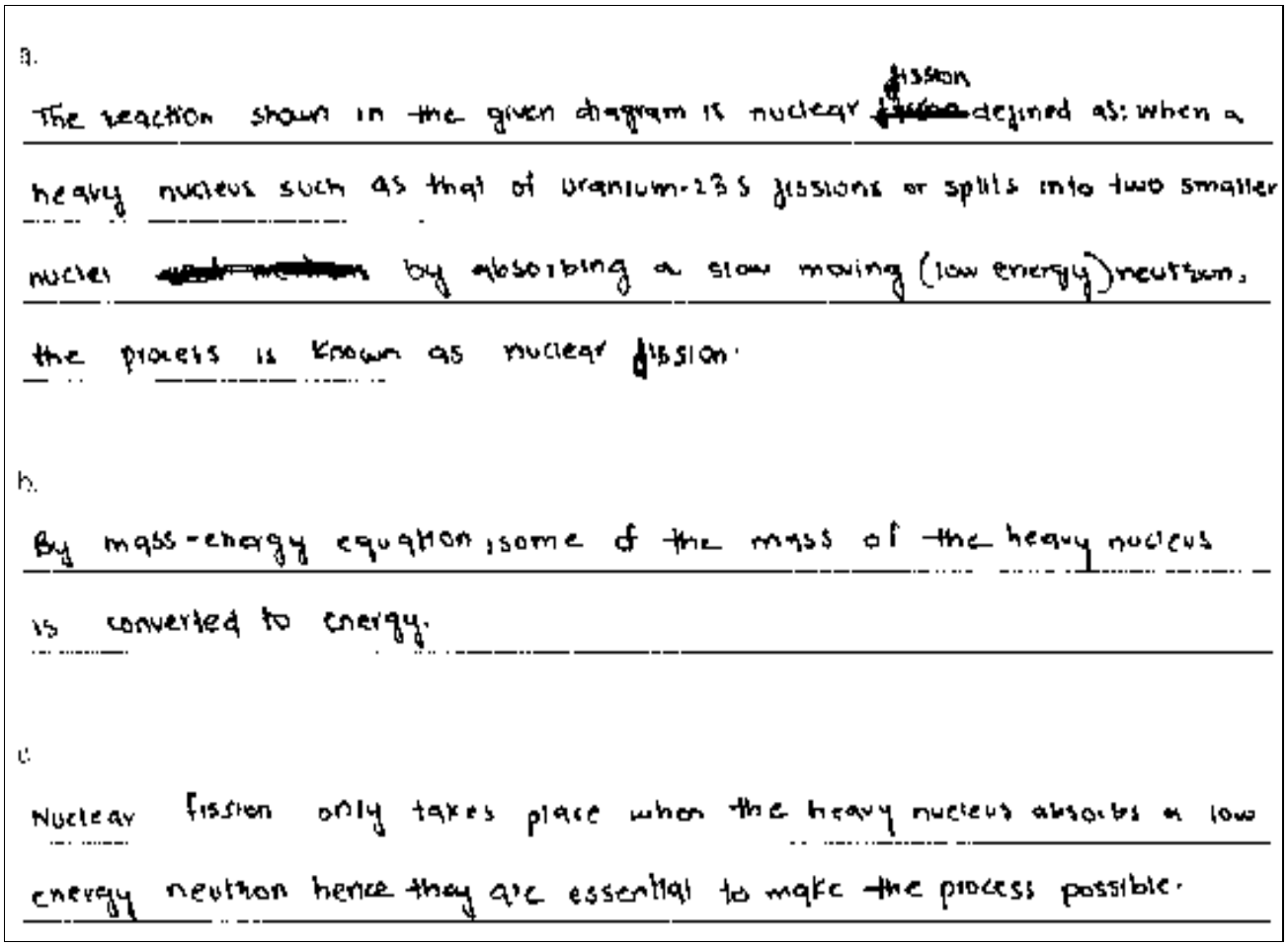
The given diagram shows a nuclear reaction.



a. Name and define the reaction shown in the diagram.
b. How can we get energy from the given process?
c. What is the role of neutrons in the given process?

Better responses correctly named and defined the type of nuclear reaction shown in the given diagram. In part (a), the name included fission chain reaction which was defined as the process in which breakdown of a large nucleus into two smaller nuclei takes place by an incident neutron. In part (b), some of the candidates gave the reference of mass and energy that are interconvertable according to $E = mc^2$ in which large amount of energy is released, whereas in part (c), candidates wrote the role of neutron as a neutral particle that approached a positively charged heavy nucleus making it heavier and unstable. Thus, the nucleus disintegrates and large amount of energy is released.

Example:

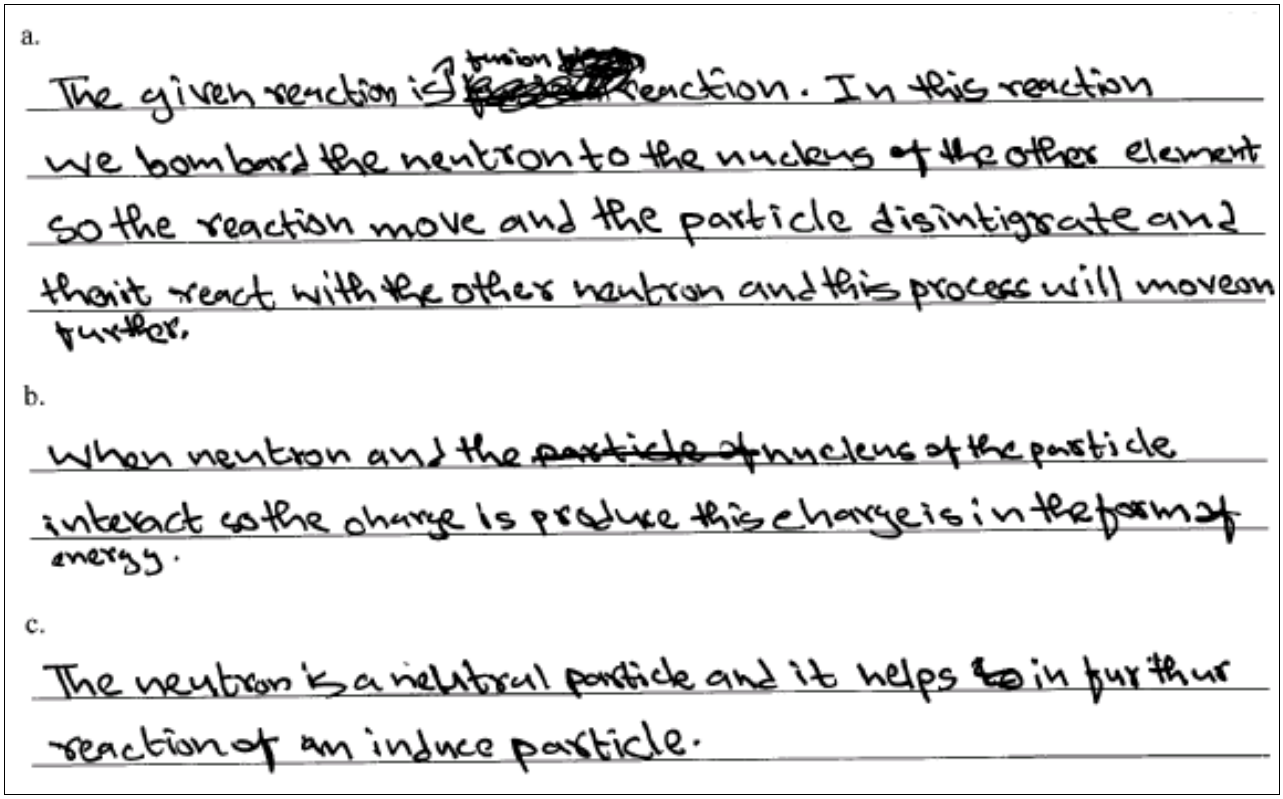

a.
The reaction shown in the given diagram is nuclear fission defined as: when a heavy nucleus such as that of uranium-235 fissions or splits into two smaller nuclei by absorbing a slow moving (low energy) neutron, the process is known as nuclear fission.

b.
By mass-energy equation, some of the mass of the heavy nucleus is converted to energy.

c.
Nuclear fission only takes place when the heavy nucleus absorbs a low energy neutron hence they are essential to make the process possible.

Weaker responses wrote incorrect answers that included background radiation, nuclear transmutation and nuclear decay. in the first part of the question. In part (b), low-scoring candidates wrote about nuclear transmutation, parent and daughter elements, stable and unstable nucleus which were completely irrelevant to the demand of the question. Some of the low-scoring candidates failed to describe neutron as a neutral particle and when nucleus disintegrates energy is released.

Example:



a.

The given reaction is fusion [illegible] reaction. In this reaction we bombard the neutron to the nucleus of the other element so the reaction move and the particle disintigrate and their react with the other neutron and this process will move on further.

b.

When neutron and the ~~particle of~~ nucleus of the particle interact so the charge is produce this charge is in the form of energy.

c.

The neutron is a neutral particle and it helps ~~to~~ in further reaction of an induce particle.

Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**

Majority of the candidates attempted question **9a** and they scored well in this part.

Question 9a:

A simple pendulum is attached with 4 m long string.

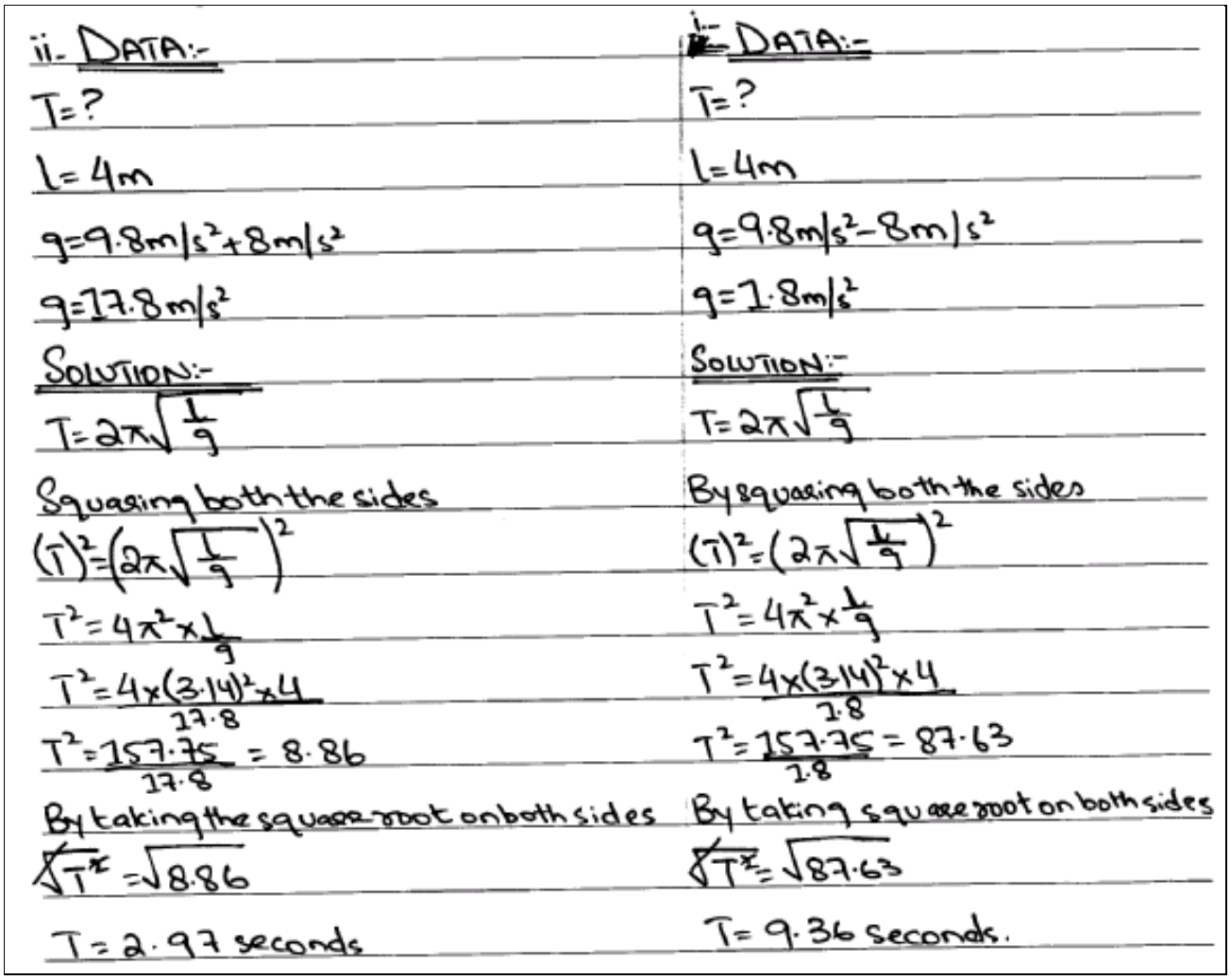
a. Find the time period of the simple pendulum if it is placed in an elevator moving upwards with an acceleration of 8 m/s^2.

b. If the elevator is moving downwards with an acceleration as mentioned of 8 m/s^2, then what will be its time period?

(**Note**: The value of acceleration due to gravity 'g' is 9.8 m/s^2.)

Better responses correctly found the time period of the simple pendulum when it is placed in an elevator moving upwards with an acceleration of 8 m/s^2. High-scoring candidates wrote the correct formula and calculated time period using $T = 2\pi\sqrt{\frac{l}{g}}$ in both cases.

Example:



ii- DATA:-

$T = ?$

$l = 4m$

$g = 9.8 m/s^2 + 8 m/s^2$

$g = 17.8 m/s^2$

SOLUTION:-

$T = 2\pi\sqrt{\frac{l}{g}}$

Squaring both the sides

$(T)^2 = \left(2\pi\sqrt{\frac{l}{g}}\right)^2$

$T^2 = 4\pi^2 \times \frac{l}{g}$

$T^2 = \frac{4 \times (3.14)^2 \times 4}{17.8}$

$T^2 = \frac{157.75}{17.8} = 8.86$

By taking the square root on both sides

$\sqrt{T^2} = \sqrt{8.86}$

$T = 2.97$ seconds

i- DATA:-

$T = ?$

$l = 4m$

$g = 9.8 m/s^2 - 8 m/s^2$

$g = 1.8 m/s^2$

SOLUTION:-

$T = 2\pi\sqrt{\frac{l}{g}}$

By squaring both the sides

$(T)^2 = \left(2\pi\sqrt{\frac{l}{g}}\right)^2$

$T^2 = 4\pi^2 \times \frac{l}{g}$

$T^2 = \frac{4 \times (3.14)^2 \times 4}{1.8}$

$T^2 = \frac{157.75}{1.8} = 87.63$

By taking square root on both sides

$\sqrt{T^2} = \sqrt{87.63}$

$T = 9.36$ seconds.

Weaker responses revealed the use of wrong formulae, i.e. S = vt which was totally irrelevant to the demand of the question. Some of the low-scoring candidates also used the equation of motion for calculating the time period of the simple pendulum.

Example:

(i) • DATA:	(ii) • DATA:
Distance = S = 4 m	Distance = S = 4m
time period = t = ? (upwards)	time period = t = ? (downwards)
acceleration = Va = 8m/s²	acceleration = Va = ?
• FORMULA:	• FORMULA:
S = Vt	S = Vt
• SOLUTION:	• SOLUTION:
4 = 8 x t	t = -0·5 s (It will take the
1 4 = t	same time to move downwards)
2 8	• RESULT:
t = 0·5 s	The time will be -0·5 s.
• RESULT:	
The time period = 0·5 s.	

Question 9b:

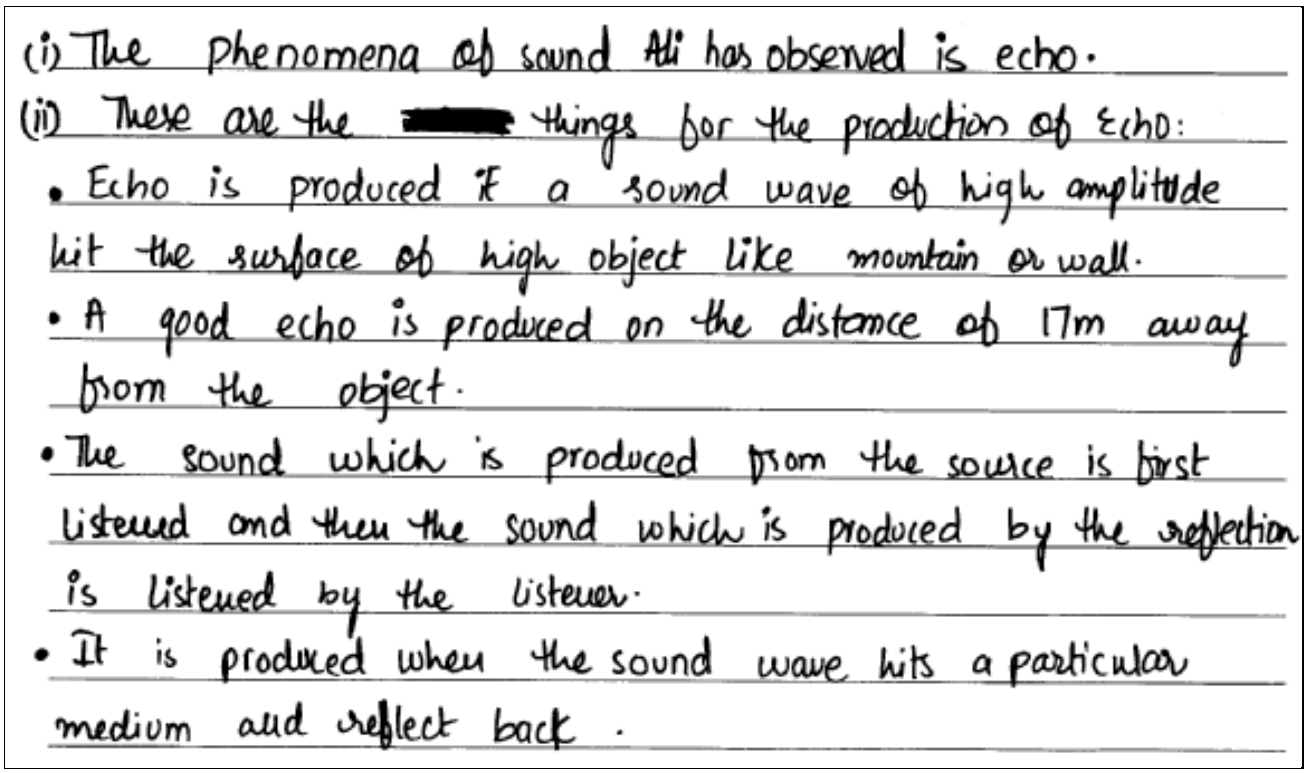
Ali produces a sound in a big empty hall. He hears a recurring sound resembling to his original sound.

i. Name the phenomenon of the sound wave that Ali observed.

ii. Explain how it is produced in FOUR points.

In better responses, majority of the candidates correctly defined 'echo' by stating that when sound incidents on the surface of a medium it bounces back into the first medium. They further elaborated that when an empty room starts to get filled with people, the echo in the room decreases. They also wrote that sound is a mechanical wave that radiated in all directions from a source. Some of the candidates also wrote that echo reflected from a flat and large surface and for the production of echo, the distance between the source and the flat reflective surface must be at least 17 m or more.

Example:

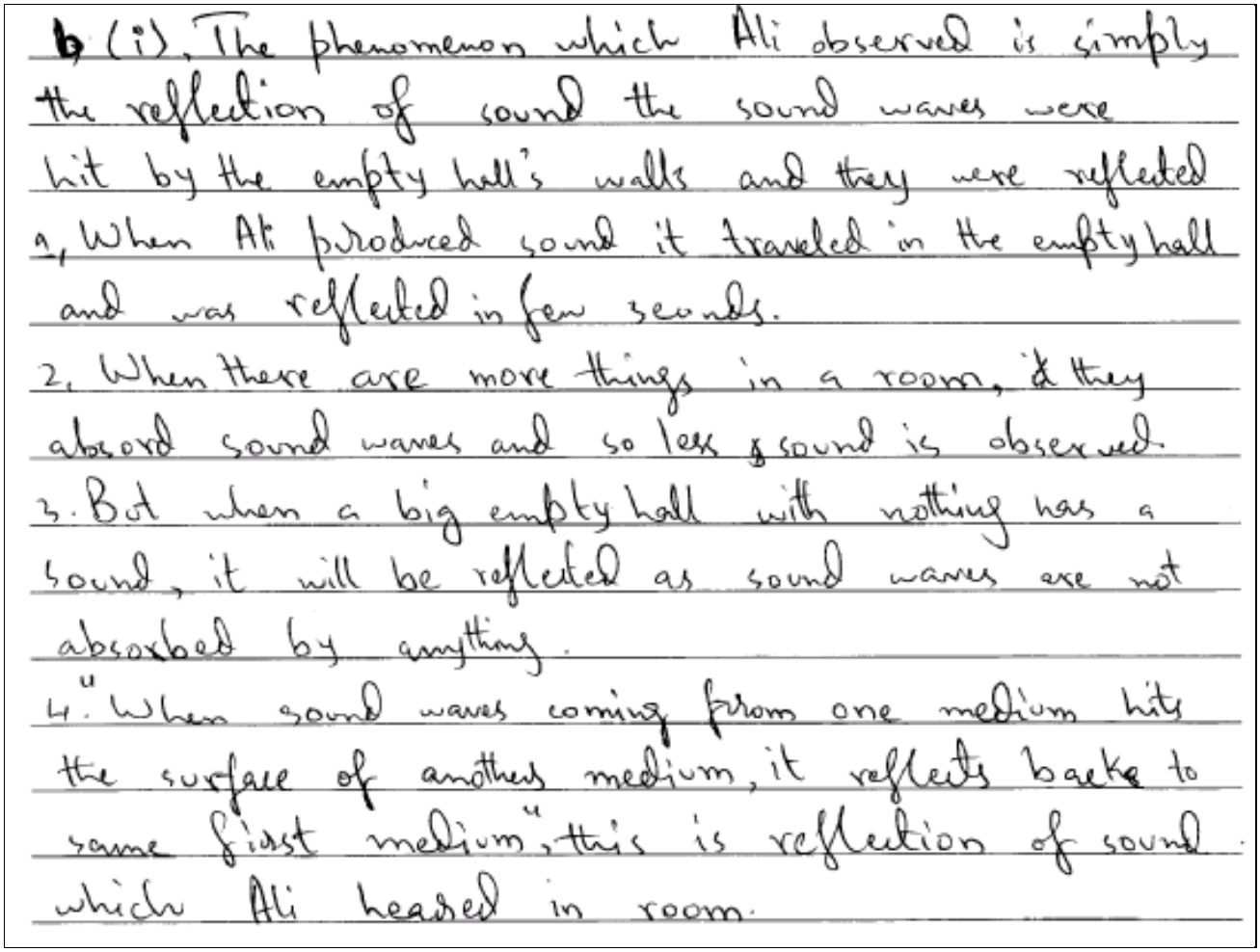

(i) The phenomena of sound Ali has observed is echo.

(ii) These are the [redacted] things for the production of Echo:

- Echo is produced if a sound wave of high amplitude hit the surface of high object like mountain or wall.
- A good echo is produced on the distance of 17m away from the object.
- The sound which is produced from the source is first listened and then the sound which is produced by the reflection is listened by the listener.
- It is produced when the sound wave hits a particular medium and reflect back.

Weaker responses reflected that candidates lack understanding of 'echo'. In the first part, they wrote about noise and pleasant sounds for human and other living organisms. In the second part of the question, low-scoring candidates were also unable to elaborate the statement about production of echo. They wrote about reflection of sound waves, volume, mass and density in their answers which were completely irrelevant to the demand of the question.

Example:



b (i). The phenomenon which Ali observed is simply the reflection of sound the sound waves were hit by the empty hall's walls and they were reflected
1, When Ali produced sound it traveled in the empty hall and was reflected in few seconds.
2, When there are more things in a room, it they absorb sound waves and so less sound is observed.
3. But when a big empty hall with nothing has a sound, it will be reflected as sound waves are not absorbed by anything.
4. When sound waves coming from one medium hits the surface of another medium, it reflects backs to same first medium, this is reflection of sound which Ali heared in room.

Majority of the candidates attempted question **10a** and they scored well in this part.

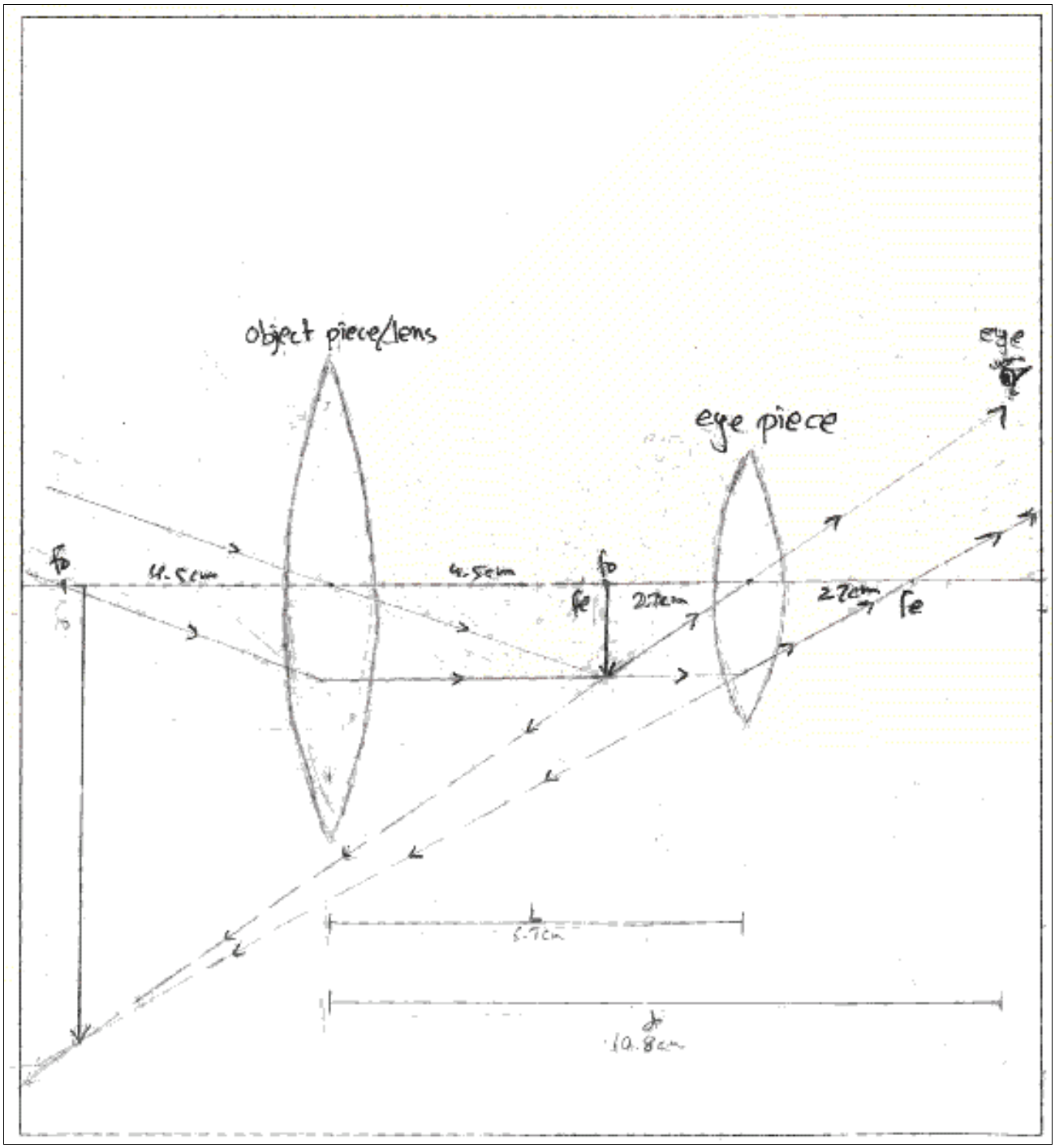
Question 10a:

Telescope is an optical instrument that uses two different lenses named as objective and eyepiece.

To obtain clear and brighter image of the distant astronomical objects, a bigger objective lens as compared to the eyepiece is used in the telescope. Draw a labelled ray diagram of this type of working telescope.

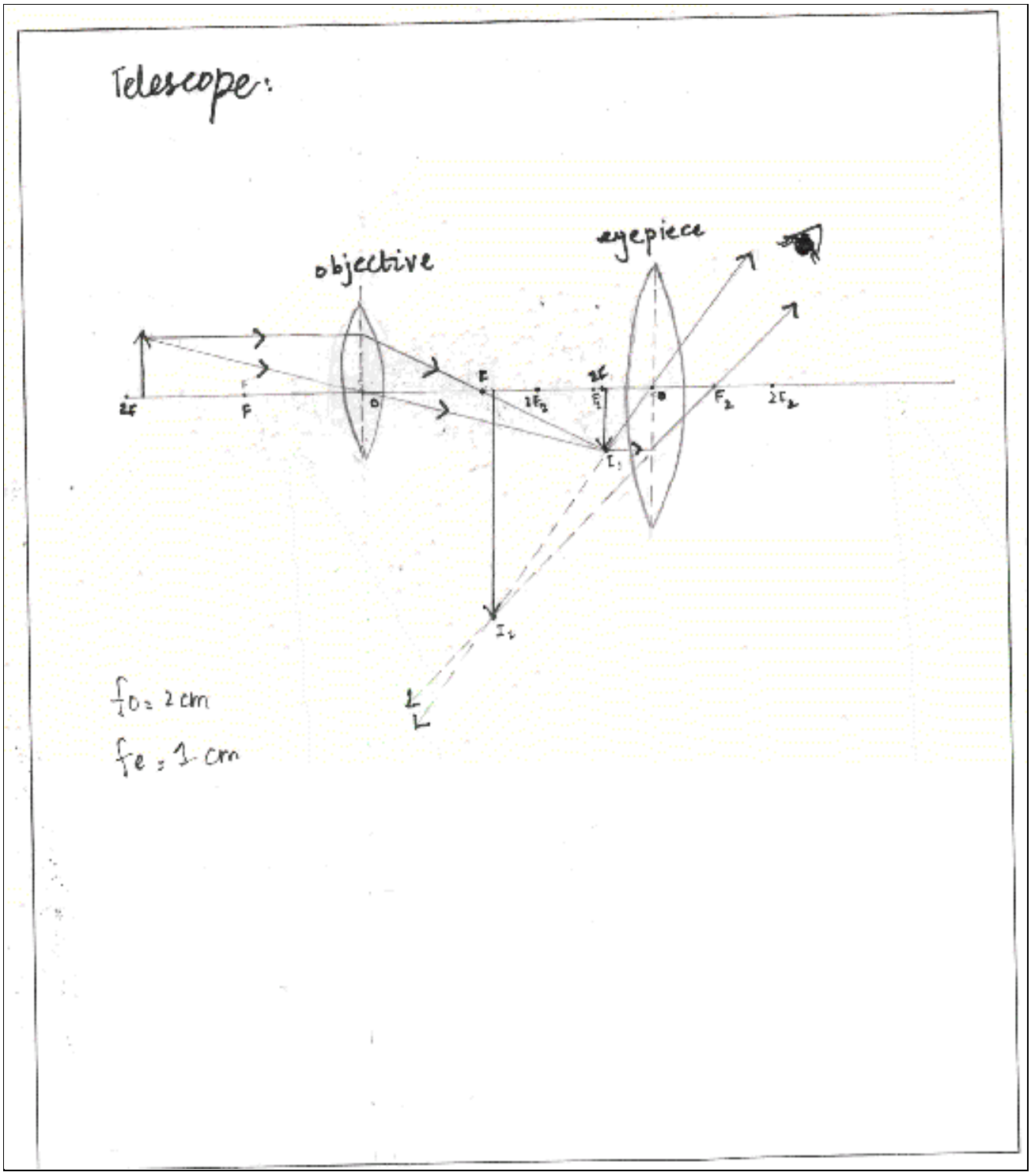
Better responses correctly drew a well labelled ray diagram of a working telescope that illustrated the formation of an enlarged and virtual image through a converging lens.

Example:



Weaker responses failed to draw a well labelled ray diagram. It seemed candidates did not know the basic difference between a compound microscope and a telescope and were unable to place the object at the right position in front of the convex lens, position of the objective and eye piece and thus, failed to draw the image as demanded in the question.

Example:



Question 10b:

An old man who can read sign boards on roads struggle to read a newspaper or a book.

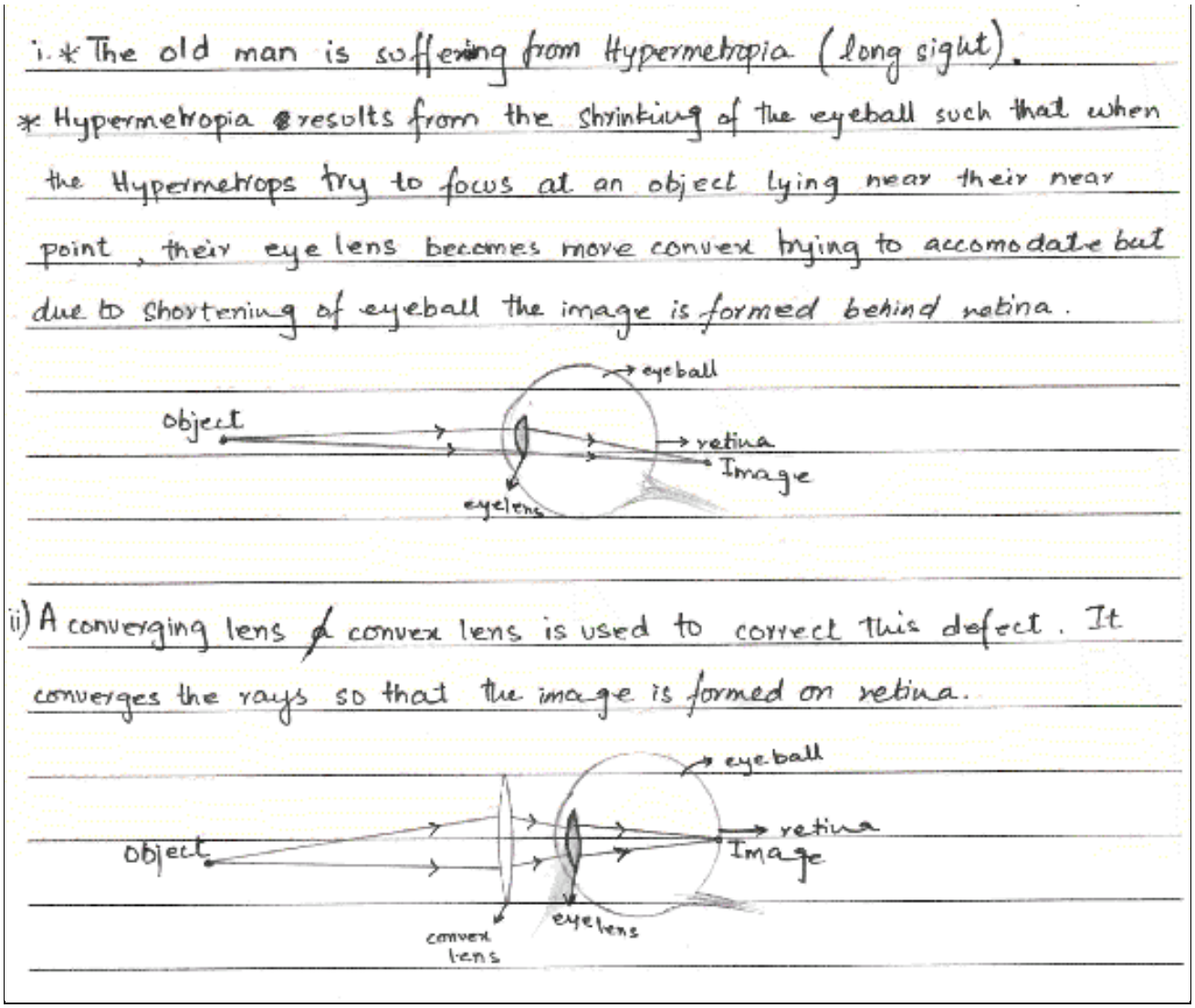
a. Identify the defect in his vision and write its cause. Also, show this defect with the help of a ray diagram.

b. Name the lens that can correct this defect. Draw a ray diagram to support your answer.

c.

Better responses correctly identified the defect in vision and wrote its cause with the help of correct ray diagram. High-scoring candidates also named the lens that can correct this defect and drew the ray diagram.

Example:

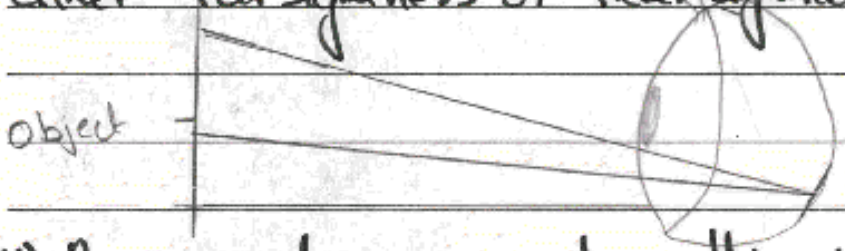


Weaker responses failed to identify the defect in vision and were unable to write its cause with the help of correct ray diagram. Some of the candidates wrote myopia and hyperopia and other defects of human eye which were totally irrelevant to the demand of the question. They were also unable to draw the correct labelled diagram.

Example:

b)

i) Myopia – nearsightedness. It happen when the image formation does not form at its original place or is either formed before the retina or after the retina. These issues cause the person to have trouble seeing clearly. This can be in both cases either farsightness or nearsightness.



ii) A convex lens can clear this issue.

Majority of the candidates attempted question **11a** and they scored well in this part.

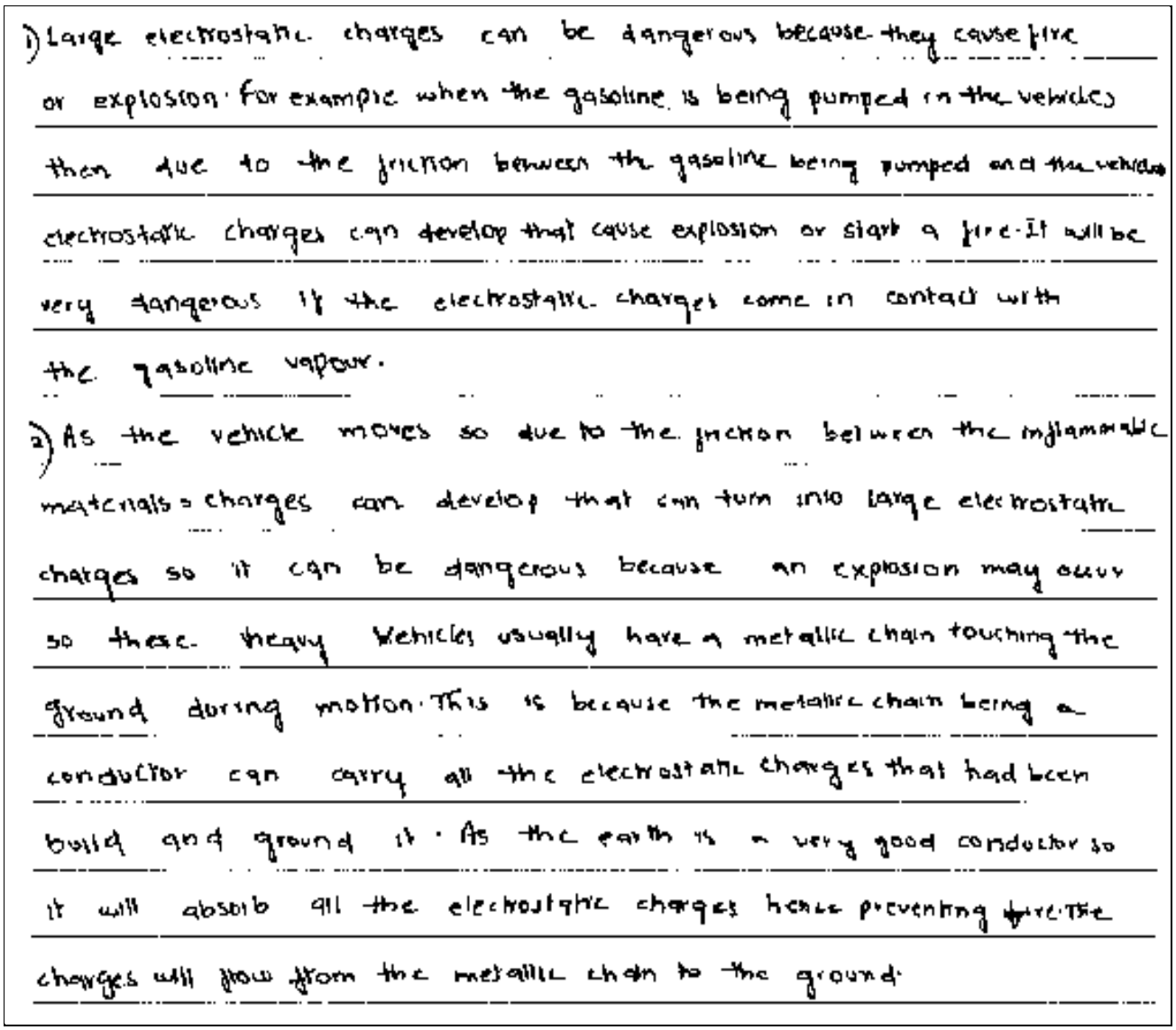
Question 11a:

a. Why large electrostatic charges can be dangerous? Give ONE example of it from daily life.

b. Long vehicles carrying inflammable materials usually have a metallic chain touching the ground during motion. Describe this statement using the phenomenon of electrostatic charges and friction.

Better responses correctly reasoned out that large electrostatic charges can be dangerous with an example from daily life. In part (a), high-scoring candidates wrote that when a large electrostatic charge builds up, it can create a spark and while refuelling vehicles this spark can ignite the fuel causing a fire. In part (b), candidates described that when a vehicle moves, its body gets charged on account of friction due to air. Some of the candidates wrote that the metallic ropes from the vehicle touching the ground enable the accumulated charges that will flow to the Earth.

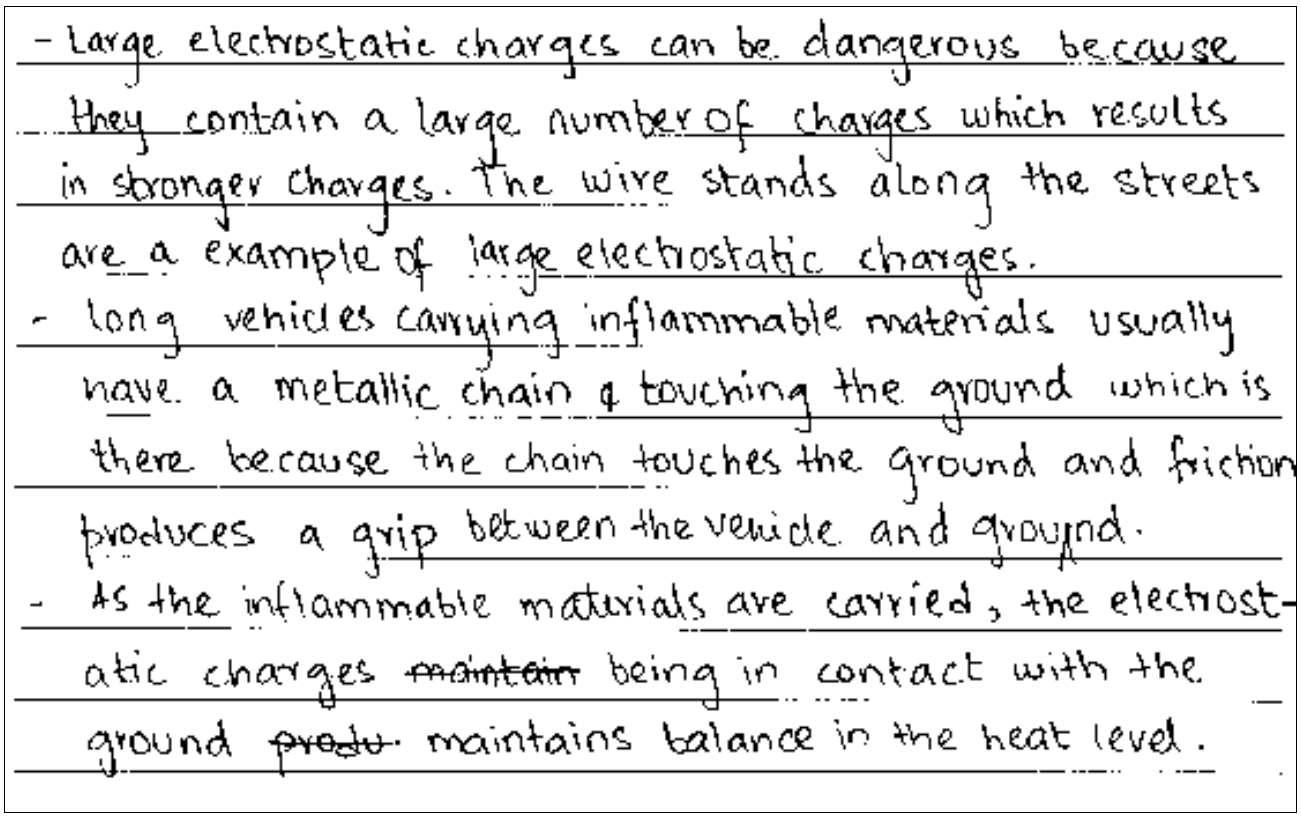
Example:


1) Large electrostatic charges can be dangerous because they cause fire or explosion. For example when the gasoline is being pumped in the vehicles then due to the friction between the gasoline being pumped and the vehicles electrostatic charges can develop that cause explosion or start a fire. It will be very dangerous if the electrostatic charges come in contact with the gasoline vapour.

2) As the vehicle moves so due to the friction between the inflammable materials charges can develop that can turn into large electrostatic charges so it can be dangerous because an explosion may occur so these heavy vehicles usually have a metallic chain touching the ground during motion. This is because the metallic chain being a conductor can carry all the electrostatic charges that had been build and ground it. As the earth is a very good conductor so it will absorb all the electrostatic charges hence preventing fire. The charges will flow from the metallic chain to the ground.

Weaker responses were unable to justify their answers and failed to describe that long vehicles carrying inflammable materials usually have a metallic chain touching the ground during motion. Some of the low-scoring candidates tried to describe the terminology used in the electrostatic induction such as electrostatic charges and earthing the overflow of charges created during the motion of vehicles.

Example:

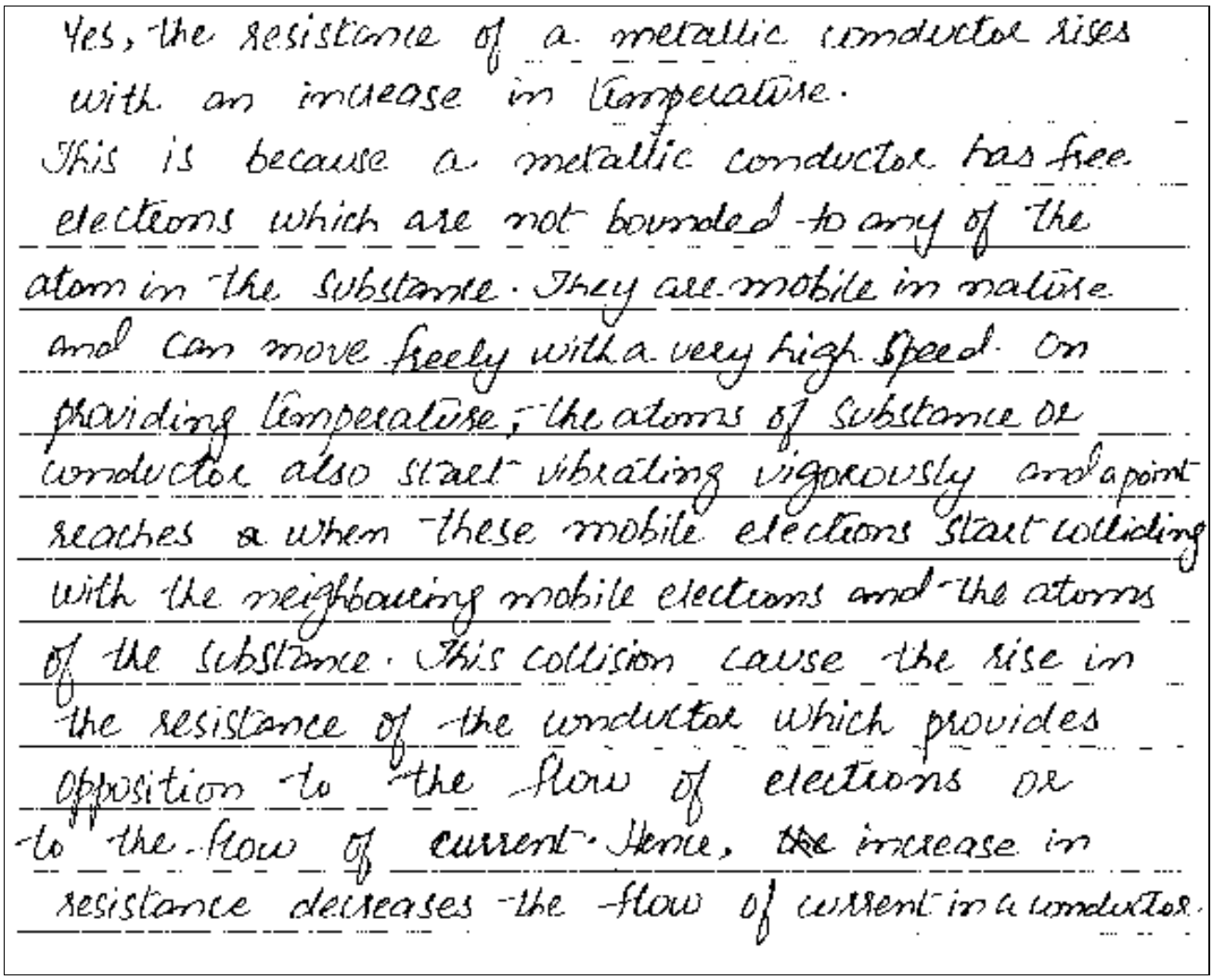

- Large electrostatic charges can be dangerous because they contain a large number of charges which results in stronger charges. The wire stands along the streets are a example of large electrostatic charges.
- long vehicles carrying inflammable materials usually have a metallic chain q touching the ground which is there because the chain touches the ground and friction produces a grip between the vehicle and ground.
- As the inflammable materials are carried, the electrostatic charges ~~maintain~~ being in contact with the ground ~~produ~~ maintains balance in the heat level.

Question 11b:

Does the resistance of a metallic conductor rise with an increase in temperature? Explain your answer with suitable reasons.

Better responses correctly wrote the reason that resistance of a metallic conductor rise with an increase in temperature. High-scoring candidates explained that the resistance offered by a metallic conductor to the electric current is due to the collisions of free electrons with the atoms present in the conductor and as well as with the rise of temperature. Some of the candidates wrote that atoms vibrate rapidly with larger amplitude and the chances of their collision with free electrons increases with the increase in temperature.

Example:


Yes, the resistance of a metallic conductor rises with an increase in temperature.
This is because a metallic conductor has free electrons which are not bounded to any of the atom in the substance. They are mobile in nature and can move freely with a very high speed. On providing temperature, the atoms of substance or conductor also start vibrating vigorously and a point reaches a when these mobile electrons start colliding with the neighbouring mobile electrons and the atoms of the substance. This collision cause the rise in the resistance of the conductor which provides opposition to the flow of electrons or to the flow of current. Hence, the increase in resistance decreases the flow of current in a conductor.

Weaker responses failed to describe in detail the concept behind the resistance of a metallic conductor that rise with an increase in temperature. Some of the low-scoring candidates reflected the relationship between temperature and resistance as directly proportional, so when the temperature rises the flow of charges in a metallic conductor felt resistance and ultimately movement of charges slows down.

Example:

Resistance of a metallic conductor rise with an increase in temperature, as more will be the temperature more will be the resistance.
When temperature increases, the metallic conductor becomes hot and the current cannot travel which increases the resistance of the metallic conductor. Temperature is directly propotional to Resistance. more ~~will~~ is be the temperature, more will be the resistance.